Customer Shopping Behavior & Transaction Analysis

Tech Stack: Python (Pandas, NumPy), MySQL, Power BI, ETL, Database Integration

- **End-to-End Data Pipeline:** Developed a complete data workflow, extracting and cleaning transactional data of 3,900 purchases across 18 features using Python, and establishing a seamless connection to load processed data into a MySQL database for structured querying.
- **Data Cleansing & Feature Engineering:** Handled missing data by executing category-specific median imputation for review ratings, engineered new tracking features (age_group, purchase_frequency_days), standardized schemas to snake_case, and eliminated redundant indicators to optimize database storage.
- **Advanced SQL Analytics:** Architected complex relational queries to extract critical business metrics, identifying a massive loyal customer segment (3,116 users), calculating standard vs. express shipping spend variances, and uncovering gender-based revenue distribution (Male: $157.8K vs. Female: $75.1K).
- **Behavioral & Promotional Insights:** Discovered that 27% of the user base drives $62.6K in subscription revenue, evaluated high-spending discount users across 839 accounts, and isolated products with extreme discount dependency (e.g., Hats at 50% discount rate) to protect profit margins.
- **Interactive Power BI Dashboard:** Built a dynamic, multi-filter executive dashboard tracking key performance indicators (KPIs) such as total customer volume (3.9K), average purchase amount ($59.76), and average rating (3.75), visualizing category sales and age group contributions to drive target marketing strategies.